state
costate
ρ
dρ
θ
dθ
ϕ
dϕ
✓ adnlp-ipopt
✓ exa-ipopt
✓ adnlp-madnlp
✓ exa-madnlp
✓ jump-ipopt
✓ jump-madnlp
1.36862250×10^0
1.36862200×10^0
1.36862150×10^0
1.36862100×10^0
1.36862050×10^0
t
control
un
uθ
uϕ
t